Core Network Technologies

1) **Network Slicing:** It is a 5G technology that allows a single physical network to be divided into multiple virtual networks (called network slices). Each slice is customized to meet eh specific requirements of different applications or industries.

Instead of building separate physical networks, operators can create multiple logical networks on the same infrastructure.

Advantages:

- Efficient use of network resources
- Customized Quality of Service (QoS)
- Better security and isolation
- Supports multiple services simultaneously
- Lower operational cost

Types of Network Slices:

A) **EMBB (Enhanced Mobile Broadband):** eMBB is designed for applications requiring very high bandwidth and high data speeds.

Characteristics:

- High download and upload speed
- High network capacity
- Supports HD and 4K video streaming

Applications:

- Video Streaming

- Virtual Reality
- Augmented Reality
- Cloud Gaming
- Video Conferencing

B) **URLLC (Ultra-Reliable Low-Latency Communications):** URLLC is designed for applications that require extremely low latency and high reliability.

Characteristics:

- Latency below 1ms
- Very high reliability (99.999%)
- Fast response time

Applications:

- Autonomous Vehicles
- Remote Surgery
- Industrial Automation
- Smart Manufacturing
- Robotics

C) **MMTC (Massive Machine Type Communications):** mMTC supports millions of connected IoT devices with low power consumption.

Characteristics:

- Supports a massive number of devices
- Low power usage
- Low data rates
- Long battery life

Applications:

- Smart Cities
- Smart Agriculture
- Environmental Monitoring
- Utility Metering
- Smart Homes

2) **Multi-access Edge Computing (MEC):** MEC brings computing and storage resources closer to end users by processing data at the edge of the network instead of sending everything to a centralized cloud. This significantly reduces communication delays and improves application performance.

Advantages:

- Lower latency
- Faster response time
- Reduced network congestion
- Better security
- Improved Quality of Experience (QoE)

Applications:

- Autonomous Vehicles
- Smart Manufacturing
- Online gaming
- Augmented Reality (AR)
- Video Analytics

Architecture:

Cloud Data Center

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Core Network (5GC)

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UPF

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Edge Server

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gNB

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UE

3) **Open Radio Access Network (O-RAN):** O-RAN is an architecture that allows equipment from different vendors to work together through standardized, open interfaces. Unlike traditional RANs that rely on a single vendor, O-RAN enables interoperability, flexibility, and innovation.

Advantages:

- Vector interoperability
- Reduced deployment cost
- Easy network upgrades
- AI-driven optimization
- Flexible network architecture

Components:

- **O-CU (Open Central Unit):** handles higher layer protocols and centralized processing
- **O-DU (Open Distributed Unit):** manages real-time radio processing
- **O-RU (Open Radio Unit):** contains radio hardware
- **Near RT RIC (Near Real-Time RAN Intelligent Controller):** performs AI-assisted optimization in near real time (typically 10 ms to 1 second)
- **Non-RT RIC (Non-Real-Time RAN Intelligent Controller):** operates over longer timescales (greater than 1 second)